Thermal Equilibrium – when objects have been in contact and the macroscopic coordinate (variables) have stopped changing.

→ involves an exchange of energy between two objects, or an object + its surroundings

→ volume

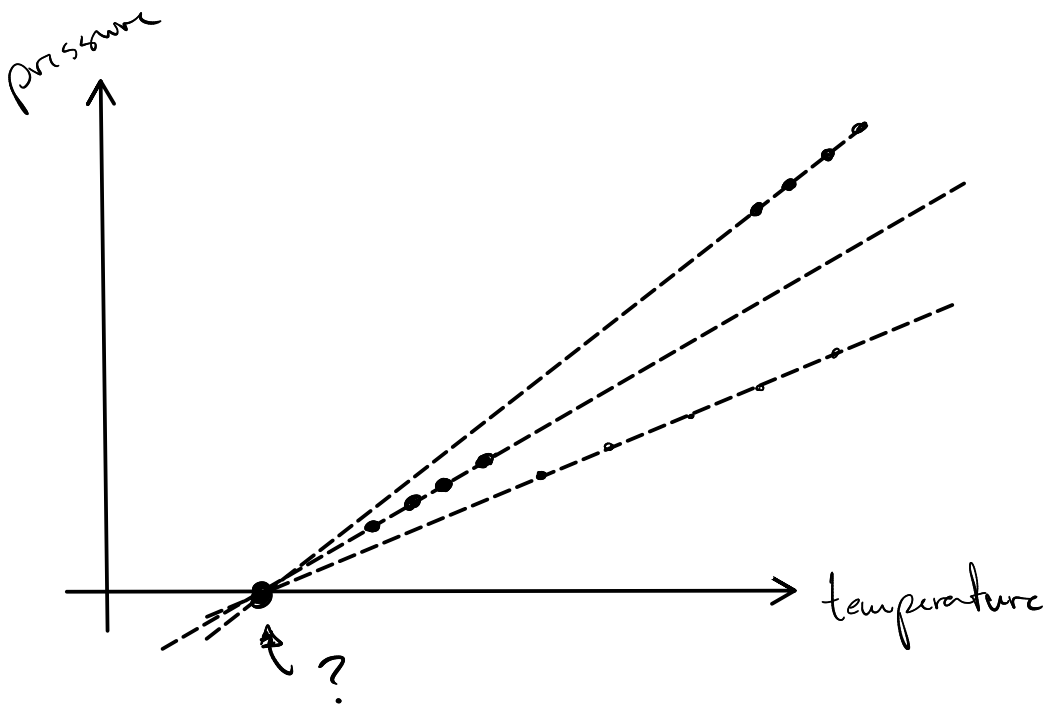
- pressure
- electrical resistance
- radiation / thermal emf

1] $F_1 = 32°F$ $F_2 = 212°F$

$C_1 = 0°C$ $C_2 = 100°C$

↳ $C = mF + b$

m: ? b: ?



Equipartition Theorem

$$U_{thermal} = N \cdot f \cdot \frac{1}{2} k_B T$$

$f = 3$ ← monatomic gas

$f = 5$ ← diatomic gas

$$N k_B = n R$$

$$U = n \cdot f \cdot \frac{1}{2} R T$$

↳ $U = \frac{3}{2} n R T$ (monatomic)

↳ $U = \frac{5}{2} n R T$ (diatomic)

Kinetic Energy

$$E_k = \frac{1}{2} m v^2$$

$$\vec{v} = v_x \hat{x} + v_y \hat{y} + v_z \hat{z}$$

$$E_k = \frac{1}{2} m \vec{v} \cdot \vec{v}$$

$$E_k = \frac{1}{2} m \left(v_x^2 + v_y^2 + v_z^2 \right)$$

Spring Potential

$$U_{sx} = \frac{1}{2} k_x x^2$$

$$U_{sy} = \frac{1}{2} k_y \cdot y^2$$

$$U_{sz} = \frac{1}{2} k_z z^2$$

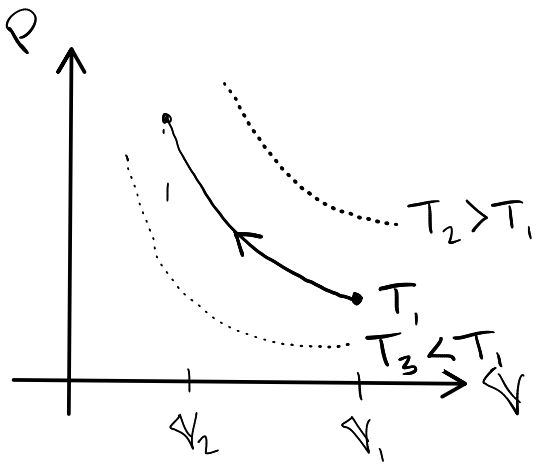
Rotation

$$K_{rot} = \frac{1}{2} I \omega^2$$

Isobaric (constant pressure) + Isochoric (constant volume), we have discussed in worksheet (week3)

Isothermal (constant temperature)

↳ compression



$$PV = \underbrace{Nk_BT}_{\text{constant!}}$$

$$P = \frac{Nk_BT}{V} \leftrightarrow \text{hyperbola}$$

$$W = -\int P\,dV$$

$$= -Nk_BT \int_{V_1}^{V_2} \frac{1}{V}\,dV$$

$$\boxed{W = -Nk_BT \ln\left(\frac{V_2}{V_1}\right)} \rightarrow W = Nk_BT \ln\left(\frac{V_1}{V_2}\right)$$

so what about the heat?

first law $\Delta U = Q + W$

$$\Delta U = \frac{f}{2} N k_B \cancel{\Delta T}^{\,0}$$

$$Q = -W$$

$$Q = Nk_BT \ln\left(\frac{V_2}{V_1}\right)$$

Adiabatic Process (isentropic) (no heat added or removed)

↳ compression

first law: $\Delta U = \cancel{Q}^{0} + W$

$$\boxed{\frac{f}{2} N k_B \Delta T = W}$$

$PV = Nk_BT$

$P = \frac{Nk_BT}{V}$

$W = \int P\, dV$

$y = -\frac{dy}{dx}$

↳ $y =$ ~

$dU = \cancel{dQ}^{0} + dW$

$dU = dW$

$\frac{f}{2} N k_B dT = -P\, dV$

$P = \frac{Nk_BT}{V}$

$\frac{f}{2} \cancel{N} k_B dT = -\frac{\cancel{N} k_B T}{V} dV$

$\frac{f}{2} dT = -\frac{T\, dV}{V}$

$\frac{f}{2} \int_{T_1}^{T_2} \frac{dT}{T} = -\int_{V_1}^{V_2} \frac{dV}{V}$

$\frac{f}{2} \ln\left(\frac{T_2}{T_1}\right) = -\ln\left(\frac{V_2}{V_1}\right)$

$\ln\left[\left(\frac{T_2}{T_1}\right)^{\frac{f}{2}}\right] = \ln\left(\frac{V_1}{V_2}\right)$

$$V \propto \frac{1}{T^{f/2}} \quad \longleftarrow \quad \left(\frac{T_2}{T_1}\right)^{\frac{f}{2}} = \left(\frac{V_1}{V_2}\right)$$

7] $$\beta = \frac{\Delta V / V}{\Delta T}$$

$$\beta = \frac{1}{5500\,K} = 1.81 \cdot 10^{-4}\,K^{-1}$$

(a) $(0.25\,cm)^3 =$

$(2.5\,mm)^3 = \underline{\underline{10\,mm^3}}$

$$\beta = \frac{\Delta V / V}{\Delta T}$$

$1°C \rightarrow 1\,mm$ of tube length

$1K$

$$1.81 \cdot 10^{-4}\,K^{-1} = \frac{\Delta V / 10\,mm^3}{1\,K} \Rightarrow \Delta V = 1.81 \cdot 10^{-3}$$

$$\Delta V = \pi r^2 \cdot l$$

$$1.581 \cdot 10^{-3}\ mm^3 = \pi r^2 \cdot 1\ mm$$

$$r = 0.02\ mm$$

$$\boxed{d = 0.04\ mm}$$

b] $\alpha = \dfrac{\frac{\Delta L}{L}}{\Delta T} \Rightarrow \dfrac{\Delta L}{L} = \alpha \cdot \Delta T$

$\frac{\Delta L}{L}$: does not matter which unit you use → $[K^{-1}]$

a] $\alpha = 1.1 \cdot 10^{-5}\ K^{-1}$

$T_{cold} = -10°C$

$T_{hot} = 40°C$ $\Rightarrow \Delta T = 50\ K$

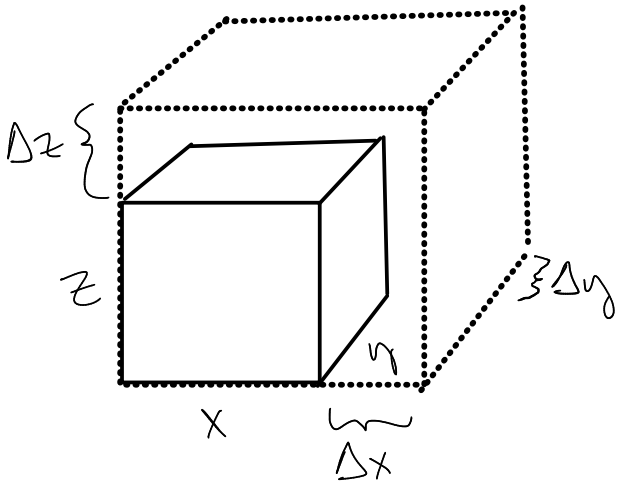
$$\Delta L = \alpha \cdot \Delta T \cdot L$$

$$= 1.1 \cdot 10^{-5} \cdot 50K \cdot 1\ km$$

$$= 55 \cdot 10^{-5}\ km \cdot \frac{1000\ m}{1\ km} = 55 \cdot 10^{-2} = 0.55\ m =$$

c)

$$\alpha \Delta T = \frac{\Delta L}{L}$$



$$\alpha_x \Delta T = \frac{\Delta x}{x} \quad \Big| \quad \alpha_y \Delta T = \frac{\Delta y}{y} \quad \Big| \quad \alpha_z \Delta T = \frac{\Delta z}{z}$$

$$\beta \Delta T = \frac{\Delta V}{V}$$ ← what is ΔV in terms of $\Delta x, \Delta y, \Delta z$

$$\Delta V = V_f - V_i = (x + \Delta x)(y + \Delta y)(z + \Delta z) - x \cdot y \cdot z$$

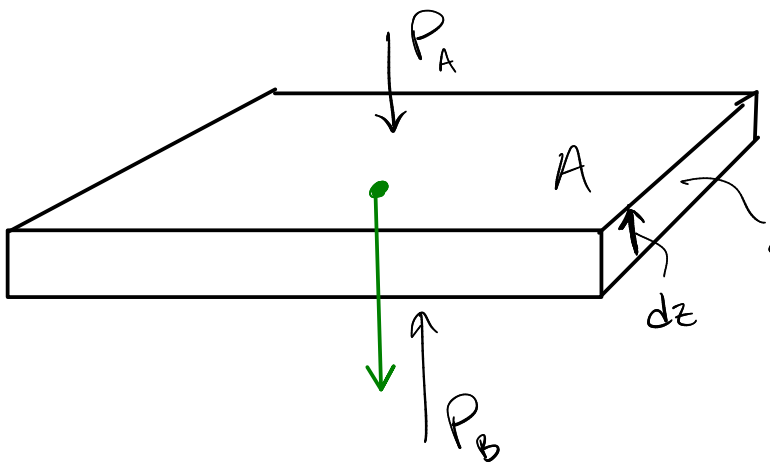
$$= (xy + y\Delta x + x\Delta y + \Delta x \Delta y)(z + \Delta z) - xyz$$

$$= \cancel{xyz} + xy\Delta z + yz\Delta x + y\Delta x\Delta z + xz\Delta y + x\Delta y\Delta z + z\Delta x\Delta y + \Delta x\Delta y\Delta z - \cancel{xyz}$$

$$\beta \Delta T = \frac{\Delta V}{V} = \frac{\Delta V}{xyz} = \underbrace{\frac{\Delta z}{z}}_{\alpha_z \Delta T} + \underbrace{\frac{\Delta x}{x}}_{\alpha_x \Delta T} + \cancel{\frac{\Delta x \Delta z}{xz}}^{\,0} + \underbrace{\frac{\Delta y}{y}}_{\alpha_y \Delta T} + \cancel{\frac{\Delta y \Delta z}{yz}}^{\,0} + \cancel{\frac{\Delta x \Delta y}{xy}}^{\,0} + \cancel{\frac{\Delta x \Delta y \Delta z}{xyz}}^{\,0}$$

$$\beta = \alpha_x + \alpha_y + \alpha_z$$

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$\rho = \frac{M}{V}$

$dm = \rho\, dV$

$= \rho A \cdot dz$

a) $\Sigma F_y = P_B \cdot A - P_A \cdot A - \rho A\, dz \cdot g = 0$

$-dP$

$-dP = \rho g\, dz$

$$\boxed{\frac{dP}{dz} = -\rho g}$$

b) $\frac{dP}{dz} = -\rho g$

$= -\frac{mN}{V} \cdot g$

$PV = Nk_BT \Rightarrow \frac{N}{V} = \frac{P}{k_BT}$

$V = \frac{Nk_BT}{P}$

$$\frac{dP}{dz} = -\underbrace{\frac{mg}{k_B T}}_{c} \cdot P$$

$$\frac{dP}{dz} = -c \cdot P$$

$$dP = -c \cdot P \cdot dz$$

$$\int \frac{dP}{P} = \int -c \, dz$$

↖ integration constant

$$\ln P = -cz + A$$

$$e^{\ln P} = e^{-cz+A} = e^{-cz} \cdot e^{A} = \underbrace{e^{A}}_{A} \cdot e^{-cz}$$

$$P = A e^{-cz}$$

$$\boxed{P(z) = A e^{-\frac{mgz}{k_B T}}} \longrightarrow P(0) = A = P_0$$

↳ sea level pressure = 1 atm = $1.013 \cdot 10^5$ Pa

1.17] a] $P = \frac{nRT}{V}\left(1 + \frac{n}{V} B\right)$

$$P = \frac{n}{V} RT + \left(\frac{n}{V}\right)^2 \cdot BRT$$

unknown

x

$$P = x \cdot RT + x^2 \cdot BRT$$

$$0 = BRT \cdot x^2 + RT \cdot x - P$$

$$\frac{n}{V} = x = \frac{-RT \pm \sqrt{(RT)^2 - 4 \cdot BRT(-P)}}{2 \cdot BRT}$$

	T	B	B2	n/V+	n/V-	n/V ideal	n/V * B
0	100	-160.0	-0.000160	122.747662	6127.252338	120.336943	-0.019640
1	200	-35.0	-0.000035	60.295717	28511.132855	60.168472	-0.002110
2	300	-4.2	-0.000004	40.119075	238055.119021	40.112314	-0.000169
3	400	9.0	0.000009	30.076095	-111141.187206	30.084236	0.000271
4	500	16.9	0.000017	24.057608	-59195.655241	24.067389	0.000407
5	600	21.3	0.000021	20.047597	-46968.404404	20.056157	0.000427

c] $PV = nRT\left(1 + \left(\frac{n}{V}\right)B + \left(\frac{n}{V}\right)^2 C\right)$ ← virial

$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$ ← vDW

$\left(P + \frac{an^2}{V^2}\right) \cdot V \cdot \left(1 - \frac{nb}{V}\right) =$

$\left(P + \frac{an^2}{V^2}\right) \cdot V = nRT\left(1 - \frac{nb}{V}\right)^{-1}$

$PV + \frac{an^2}{V} =$

$PV = nRT\left(1 - \frac{nb}{V}\right)^{-1} - \frac{an^2 \cdot nRT}{V \cdot nRT}$

$= nRT\left[\left(1 - \frac{nb}{V}\right)^{-1} - \frac{an}{VRT}\right]$

$$(1+x)^p \approx 1 + px + \frac{1}{2}p(p-1)x^2$$

$$p = -1 \qquad x = -\frac{nb}{V}$$

$$= nRT\left[1 - \left(-\frac{nb}{V}\right) + \frac{1}{2}(-1)(-2)\left(-\frac{nb}{V}\right)^2 - \frac{an}{RTV}\right]$$

$$\text{vDW} \rightarrow \quad PV = nRT\left[1 + \frac{nb}{V} + \left(\frac{nb}{V}\right)^2 - \frac{an}{RTV}\right]$$

$$\frac{n}{V}\left(b - \frac{a}{RT}\right)$$

$$\text{virial} \rightarrow \quad PV = nRT\left(1 + \left(\frac{n}{V}\right)B + \left(\frac{n}{V}\right)^2 C\right)$$

$$B = b - \frac{a}{RT} \quad \checkmark$$

$$C = b^2$$

C] $B = b - \frac{a}{RT}$ $B(T)$ from data

1.36, 1.40, 1.45

1.36

(a) $V_1 = 1 \text{ liter} = .001 \text{ m}^3$ $P_1 = 1 \text{ atm} = 1 \cdot 10^5 \text{ Pa}$

$V_2 = ?$ $P_2 = 7 \text{ atm} = 7 \cdot 10^5 \text{ Pa}$

$\gamma = \frac{7}{5}$ for air

$$\frac{V_2^\gamma}{V_1^\gamma} = \frac{P_1}{P_2}$$

$$V_2 = V_1 \left(\frac{P_1}{P_2}\right)^{1/\gamma} = 1 \text{ liter} \left(\frac{1 \text{ atm}}{7 \text{ atm}}\right)^{5/7}$$

$$V_2 = .249 \text{ liters}$$

(b) work done

$dU = \pm W$

$\Delta U = W$

$\frac{5}{2} N k_B \Delta T = W$

↑ so what is final T and N?

It is easier to solve for part c. first and then to do, but the alternative way is possible w/ an integral

Alternative: $dW = -p\,dV$

$pV^\gamma = \text{constant}$

$$\frac{p_2}{p_1} = \frac{V_1^\gamma}{V_2^\gamma} \Rightarrow p(V) = \frac{p_1 V_1^\gamma}{V^\gamma}$$

$$dW = -\frac{p_1 V_1^\gamma}{V^\gamma} dV$$

$$W = -\int_{V_1}^{V_2} \frac{p_1 V_1^\gamma}{V^\gamma} dV = -p_1 V_1^\gamma \underbrace{\int_{V_1}^{V_2} \frac{dV}{V^\gamma}}_{\int_{V_1}^{V_2} V^{-\gamma} dV}$$

$$\frac{1}{-\gamma+1} V^{-\gamma+1} \Big|_{V_1}^{V_2}$$

$$-\frac{1}{\gamma-1} V^{-\gamma+1} \Big|_{V_1}^{V_2}$$

$$= \frac{p_1 V_1}{\gamma-1} V^{-\gamma+1} \Big|_{V_1}^{V_2}$$

$$W = \frac{p_1 V_1^\gamma}{\gamma-1} \left(V^{-\gamma+1}\right)\Big|_{V_1}^{V_2}$$

$$W = \frac{p_1 V_1^\gamma}{\gamma-1} \left(V_2^{-\gamma+1} - V_1^{-\gamma+1}\right)$$

$$= \frac{p_1}{\gamma-1} \left(\frac{V_1^\gamma}{V_2^{\gamma-1}} - V_1^\gamma V_1^{-\gamma+1}\right)$$

$$- \frac{p_1}{\gamma-1} \left(\frac{V_1^\gamma}{V_2^{\gamma-1}} - V_1\right)$$

$$= \frac{p_1 V_1}{\gamma-1} \left(\frac{V_1^{\gamma-1}}{V_2^{\gamma-1}} - 1\right)$$

$$= \frac{p_1 V_1}{\gamma-1} \left(\left(\frac{V_1}{V_2}\right)^{\gamma-1} - 1\right)$$

just algebra to save calculator keystrokes

$$= \frac{5 \cdot 10^5 \text{ Pa} \cdot 0.001 \text{ m}^3}{2} \left(\left(\frac{1}{0.25}\right)^{2/5} - 1\right) = 185 \text{ J}$$

$\gamma - 1 = \frac{7}{5} - \frac{5}{5} = \frac{2}{5}$

c. $VT^{5/2} = \text{const}$

$$\left(\frac{T_2}{T_1}\right)^{5/2} = \frac{V_1}{V_2}$$

$$T_2 = T_1\left(\frac{V_1}{V_2}\right)^{2/5} = 300K\left(\frac{1\,l}{.249\,l}\right)^{2/5}$$

$$= 300K\,(1.74) = 523\,K$$

$$250^\circ C$$

now back to part b)

$$\Delta T = T_2 - T_1 = 223\,K$$

$$N = \frac{P_1V_1}{k_BT_1} = \frac{10^5 \cdot .001\,m^3}{1.38\cdot10^{-23}\cdot 300K}$$

$$N = 2.41\cdot10^{22} \approx 0.040\ mol = n$$

$$Nk_BT_1 = PV = 10^2 \checkmark$$

$$\Delta U = \frac{5}{2}Nk_B\Delta T$$

$$= \frac{5}{2}Nk_B(T_2 - T_1)$$

$$= \frac{5}{2}\underbrace{Nk_BT_1}_{10^2}\underbrace{\left(\frac{T_2}{T_1} - 1\right)}_{\left(\left(\frac{V_1}{V_2}\right)^{7/5} - 1\right)}$$

$$= \frac{5}{2}\cdot10^2(0.74) = 185\,J \checkmark$$

1.40 T ↓ w/ ↓ altitude

a) show that ideal gas adiabatic expansion, T + P are related by $\frac{dT}{dP} = \frac{2}{f+2}\cdot\frac{T}{P}$

from 1.16: $\frac{dP}{dz} = -\frac{mg}{k_B T}\cdot P = -\frac{mg}{k_B}\frac{P}{T}$

$\frac{dP}{dT}\cdot\frac{dT}{dz} = -\frac{mg}{k_B}\cdot\frac{P}{T}$

↳ decreases, but how?

take this down to part (b) ↓

adiabatic expansion means

$dU = \cancel{dQ}^{\,0} + dW$

$dU = dW$

$\frac{f}{2}Nk_B dT = -P dV$

$= -P\cdot\frac{dV}{dP}\cdot dP$

↳ ?

So now plug this in here ↑

$PV = Nk_B T$

$V = \frac{Nk_B T}{P}$

So

$\frac{dV}{dP} = -\frac{Nk_B T}{P^2} + \frac{Nk_B}{P}\frac{dT}{dP}$

$\frac{f}{2}Nk_B dT = -P\left(-\frac{Nk_B T}{P^2} + \frac{Nk_B}{P}\frac{dT}{dP}\right)dP$

$\frac{f}{2}Nk_B dT = \left(\frac{Nk_B T}{P} + Nk_B\frac{dT}{dP}\right)dP$

$\frac{f}{2}\cancel{N}k_B dT = \frac{\cancel{N}k_B T}{P}dP + \cancel{N}k_B dT$

$$\frac{f}{2}dT = \frac{T}{p}dp + dT$$

$$\left(\frac{f}{2}+1\right)dT = \frac{T}{p}dp$$

$$\boxed{\frac{dT}{dP} = \frac{1}{\frac{f}{2}+1}\cdot\frac{T}{P}} \checkmark$$

oh wait multiply through by $\frac{2}{2}$

$$\frac{dT}{dP} = \frac{2}{f+2}\cdot\frac{T}{P} \checkmark\checkmark$$

ok now done.

b)

$$\frac{dP}{dT}\cdot\frac{dT}{dz} = -\frac{mg}{k_B}\cdot\frac{P}{T}$$

$$\frac{dT}{dP} = \frac{2}{f+2}\cdot\frac{T}{P} \Rightarrow \frac{dP}{dT} = \frac{f+2}{2}\cdot\frac{P}{T}$$

$$\frac{f+2}{2}\cdot\frac{P}{T}\cdot\frac{dT}{dz} = -\frac{mg}{k_B}\cdot\frac{P}{T}$$

from 1.16, this is the mass of one molecule. For air $= 29\frac{g}{mol}\cdot\frac{mol}{6.022\cdot10^{23}\,\text{particles}}\cdot\frac{kg}{10^3 g} = 4.81\cdot10^{-26}\,kg$

$$\frac{dT}{dz} = -\underbrace{\frac{mg}{k_B}\cdot\frac{2}{f+2}}_{\text{this is indeed constant and independent of } T+P} = -\frac{4.81\cdot10^{-26}\,kg\cdot9.8\frac{N}{kg}}{1.38\cdot10^{-23}\frac{J}{K}\cdot m}\cdot\frac{2}{5+2} = \frac{4.81\cdot9.8}{1.38}\cdot\frac{2}{7}\cdot10^{-3}\,\frac{K}{m} = 9.75\cdot10^{-3}\frac{K}{m}\cdot\frac{10^3 m}{km}$$

$$\approx 10\frac{K}{km} \checkmark\checkmark$$

1.45

$w = xy \quad x = yz$ ← given

a) $w = x\left(\frac{x}{z}\right)$ $\boxed{w = y^2 z}$ ← in terms of $y + z$

$\boxed{w = \frac{x^2}{z}}$ ← in terms of $x + y$

b) $\left(\frac{\partial w}{\partial x}\right)_y = y \qquad \left(\frac{\partial w}{\partial x}\right)_z = \frac{2x}{z}$

Since $y = \frac{x}{z}$

$$\left(\frac{\partial w}{\partial x}\right)_y = 2\left(\frac{\partial w}{\partial x}\right)_z$$

c) $\left(\frac{\partial w}{\partial y}\right)_x = x \qquad \left(\frac{\partial w}{\partial y}\right)_z = 2yz$

$\left(\frac{\partial w}{\partial z}\right)_x = -\frac{x^2}{z^2} \qquad \left(\frac{\partial w}{\partial z}\right)_y = y^2$